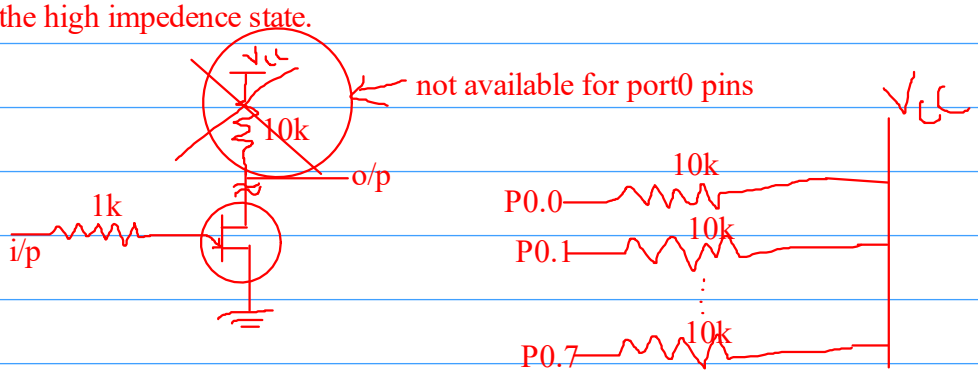
Port0 is an 8-bit open drain bi-directional I/O port.
External pullups are required during program verification.

whenever it is connected any IO device to PORT0, it is not able drive those devices.

port0 pins are having the high impedence state.



PORT1, PORT2 and PORT3 are having internal pullups.

By default all port pins are having logic high/ logic 1.

AT89C51 - Not having ISP support (need to use high voltage programmers to load the hex file)

AT89S52 - having ISP support

ISP - In-System Programming

P89V51RD2 - phillips company -- through UART hex file loading is possible with the help of flash magic software